

File Permissions in Linux (Concise Summary)

Project Overview

The research team reviewed and updated file and directory permissions in **/home/researcher2/projects** to ensure they align with organizational security requirements.

Checking Permissions

1. Navigate to the directory:

```
cd /home/researcher2/projects
```

2. Display contents and permissions:

```
ls -la
```

This command shows all files, including hidden ones, and their permission strings.

Understanding Permission Strings

- Format: **drwxrwxrwx** (10 characters)
- **1st character:** **d** for directory, **-** for file
- **User (2-4), Group (5-7), Other (8-10)** permissions
- Permissions: **r** (read), **w** (write), **x** (execute)
- Hidden files/directories begin with a dot (**.**)

Updating File Permissions

1. Remove Write Access for Others

- Issue: **project_k.txt** allowed write permission for "other" users.
- Fix:

```
chmod o-w project_k.txt  
ls -la
```

Write permissions for "other" were successfully removed.

2. Restrict Group Permissions on **project_m.txt**

- Requirement: Only the user should have **rw** permissions.

- Issue: Group had `rw` permissions.
- Fix:

```
chmod g-r project_m.txt  
ls -la
```

3. Adjust Permissions on Hidden File `.projects_x.txt`

- Issue: User had `rw`; group had `w`.
- Requirement: Both user and group should have **read-only** access.
- Fix:

```
chmod u-w,g-w .projects_x.txt  
ls -la
```

Updating Directory Permissions

`drafts` Subdirectory

- Issue: Group had execute (`x`) permission.
- Requirement: Only user should have full access.
- Fix:

```
chmod g-x /home/researcher2/projects/drafts  
ls -la
```

Summary

The project verified current permissions and corrected misconfigurations on files and directories using symbolic `chmod` operations. These updates eliminate unauthorized access and improve file system security.

Appendix: Screenshots

Below are the embedded screenshots referenced in the original document: